

L6 – Rocket Thrust Measurement

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All information documented and discussed in the following report is under the sole authorship of Anuranan Bharadwaj, Kalkamanali Satvaldy, Ryan Orr, James Nguyen. No part of this report has been published or submitted for publication.

Signed: KS, AB,
Amr, JN

Abstract

Rocket thrust testing is a fundamental part of propulsion testing. This allows engineers to describe motor performance, check manufacturer data, and understand how propellants respond under real operating settings. For this experiment, a cantilever beam equipped with a strain-gage load cell was used in this experiment to quantify thrust. The beam's bending deformation, which was sensed as a voltage change by a digital strain indicator, was calibrated to produce force. The technique began with a static calibration utilizing known masses (1 kg, 2 kg, and 3 kg) to create a linear relationship between voltage and applied load. This calibration curve was then used during the live fire testing to translate voltage-time data into thrust-time curves.

A series of commercially available Estes solid rocket motors of increasing power class (A8-3, B4-4, C6-5, C11-0, and E16-4) were placed vertically to the end of the beam, lit, and recorded using the DAQ system. The thrust profiles showed the typical characteristics of black-powder motors such as a fast decline at burnout, a sustained thrust zone, and an initial ignition surge. Important propulsion parameters including peak thrust, average thrust, burn time, and total impulse were extracted from these curves. For instance, the C6-5 motor generated a total impulse of 8.53 N·s, a recorded peak thrust of 13.55 N, an average thrust of 4.6 N, and a burn time of 2.1 s. The resulting percent deviations varied from 12% to 27% when compared to published manufacturer values (15.3 N peak, 1.6 s burn, 10 N·s total impulse), with burn time showing the most variation.

These disparities are understandable given the simplicity of the test stand and common causes of error such as ignition delays, beam dampening, modest misalignment of the motor clamp, and manufacturing tolerances in hobby-grade motors. Overall, the experiment gave practical insight into how small solid rocket motors produce thrust over time and proved the usefulness and precision of strain-gage-based thrust measurement.

Introduction

Through history humanity has found many different modes of propulsion, but none as raw and powerful as the rocket. First appearing in a recognizable form in 1232 A.D. by the Chinese to launch their arrows. These acted like solid rockets using gunpowder as a propellant [Thapa]. This technology eventually expanded into fireworks and then the Congreve rocket in the early 1800s. The first vision of the modern rocket and rocket engine can be seen in the research by Konstantin Tsiolkovsky in 1903. Tsiolkovsky was one of the first to suggest liquid propellant and provided research showing the importance of exhaust velocity in a rocket's design. The first real experimentation on this work happened in 1915 by Robert Goddard starting in 1915. Goddard used solid fuels to start measuring the exit velocities and started research on how to build a liquid fuel rocket; becoming successful in this endeavor in 1926. From here until current rockets much of the history is based upon liquid rockets but they all use the same basic principles refined from the German V2's all the way to the modern SLS. One of the most well-known examples of solid rockets would be the solid rocket boosters for the space shuttle, along with other assist boosters.

Solid rockets work of a very simple process compared to liquid rockets. Internally they consist of their propellant and an escape nozzle. The propellant consists of the fuel and oxidizer combined into one mixture called grain. This mixture is then lit from the nozzle side to create its thrust it then burns through the rocket. The cross section of the grain controls the amount of thrust produced. As the cross section can change from the start the grain has to be pushed back from the nozzle so that the entire cross section is available from the start. This grain burns continuously so in order to control the burn the geometry can be changed with various shapes causing different thrust graphs.

Both solid and liquid use their propellant to create a pressure difference inside the nozzle, along with pushing the gases out of combustion chamber into the nozzle. The gases velocity and pressure difference then work together to create the thrust. This thrust then can be used by integrating it over the burn time to find the impulse of the rocket. This impulse can be used to compare the efficiency of rockets. The rocket engines used in this experiment are solid model rocket engines by Estes. They come in many different variants but have a relatively simple designator: X#-#. In this the X represents a letter that has a total impulse designation, the first number then represents the average thrust in newtons, while the second number is the number of seconds between end of thrust and the ejection charge firing. The full designations are listed by Estes.

Equipment/Apparatus

Below is the list of equipment that was used in this experiment:

- Rolling equipment cart
- Cantilever beam
- Aluminum strip clamping block
- Damping fluid submersed damping rod/plate
- Digital Strain indicator configured in a half bridge arrangement to measure $\mu\epsilon$
- Data Acquisition Device
- Data Acquisition Software
- Personal computer and monitor
- Calibration masses
- Scissors
- Hose clamps
- Solid rocket motors, igniters and plugs
- Launch controller, including batteries, wires, and alligator clips
- Collection of wires, cables, and connectors
- Extension cord
- Safety goggles
- Outdoor tent

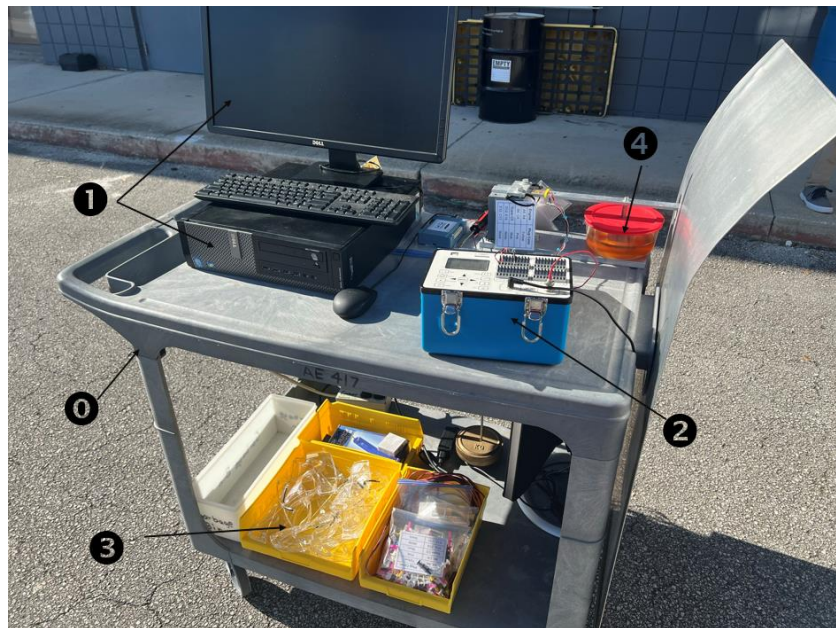


Figure 1: (0) – Rolling equipment Cart, (1) – Personal computer and monitor, (2) – Digital Strain Indicator, (3) – Safety goggles, (4) – Damping fluid (STP Oil Treatment)



Figure 2: Solid rocket plugs

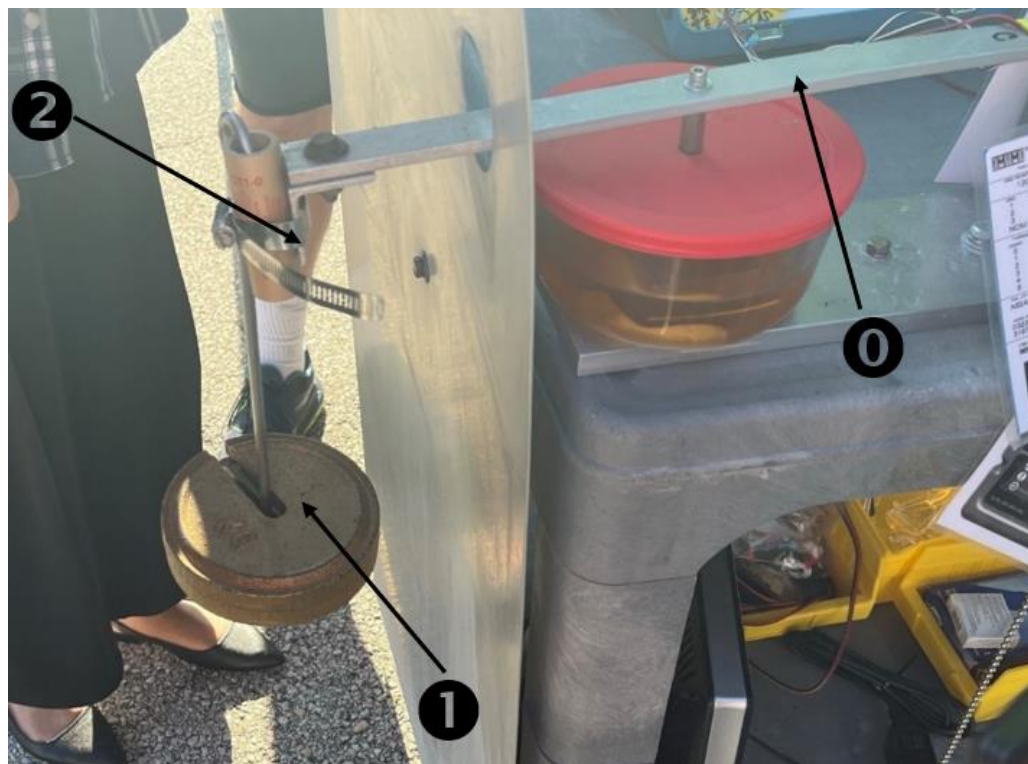


Figure 3: (0) – Cantilever beam, (1) – Calibration masses, (2) – Aluminum strip clamping block



Figure 4: Solid rocket igniters



Figure 5: Solid rocket motors

Method

A. Equipment Setup

- The equipment cart was moved outdoors to an open area with access to an electrical outlet.
- A canopy tent was set up to reduce glare on the DAQ laptop screen and to be safe from rain, if any.
- The power strip, computer, monitor, and digital strain indicator were turned on.
- The DAQ software (TracerDAQ) was opened and the preset configuration file was loaded.
- The strain indicator readings were observed to confirm they were stable before calibration began.



Figure 6: Equipment Cart moved Outdoors

B. Static Calibration of the Strain-Gage Beam

- A dummy rocket motor was attached to the end of the aluminum cantilever beam using a hose clamp.
- Data acquisition was started.
- Calibration masses of 1 kg, 2 kg, and 3 kg were placed (in order) on top of the clamped object to apply known downward forces.
- Each mass was held in place for 3 seconds to record a stable voltage plateau.
- The masses were removed and data acquisition was stopped.
- A screenshot of the graph was saved.



Figure 7: Calibration

C. Preparing the Rocket Motor

- A solid rocket motor was selected for testing.
- The igniter was inserted into the nozzle and secured using the appropriate color-coded plug. The nozzle was pointing downward.
- Two alligator clips were attached to the igniter leads while keeping the wires separated to prevent shorting.



Figure 8: Experiment Setup including Cantilever Beam

D. Mounting the Rocket Motor

- The rocket motor was positioned vertically at the end of the beam.

- The motor was tightened in place with a hose clamp so that the thrust force would act along the beam axis.
- The fire-safe container and deflector shield were aligned beneath the nozzle.

E. Firing

- Data recording was started and the system was allowed to run for approximately three seconds before ignition.
- A team member initialised the launch controller by inserting the safety key.
- The “FIRE” button was held until the motor ignited.
- The thrust vs time curve was observed on the computer as the motor produced thrust.
- Recording was stopped at burnout, and both the screenshot and CSV data file were saved.
- The motor was left to cool before being removed from the clamp.

F. Repeating the Test

- The next motor type was installed.
- Same steps from Sections C through E were repeated.
- Multiple motors were tested to compare their thrust characteristics.

G. Clean-Up

- The strain indicator, computer, and power strip were turned off.
- The canopy tent was disassembled, and the equipment cart was brought back indoors.
- All recorded data files were backed up to a USB drive.

Results

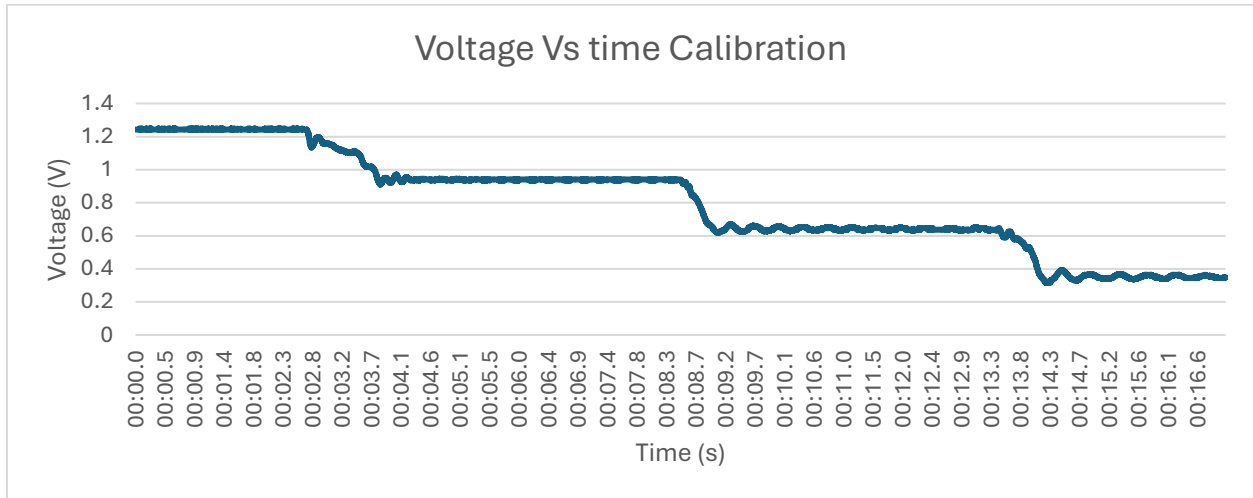


Figure 8: Voltage vs Time graph for Calibration test

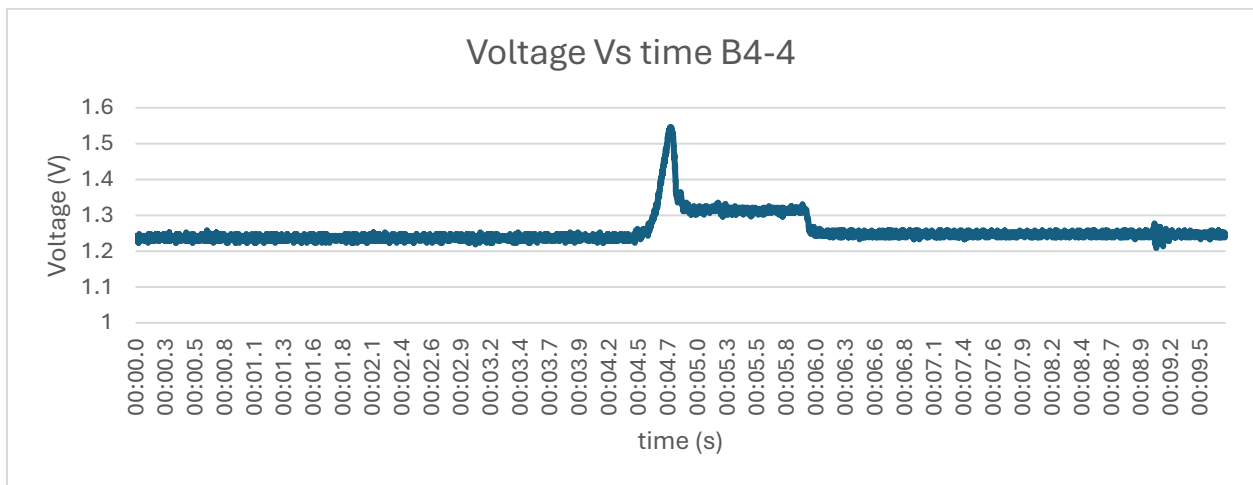


Figure 9: Voltage vs Time graph for B4-4 test

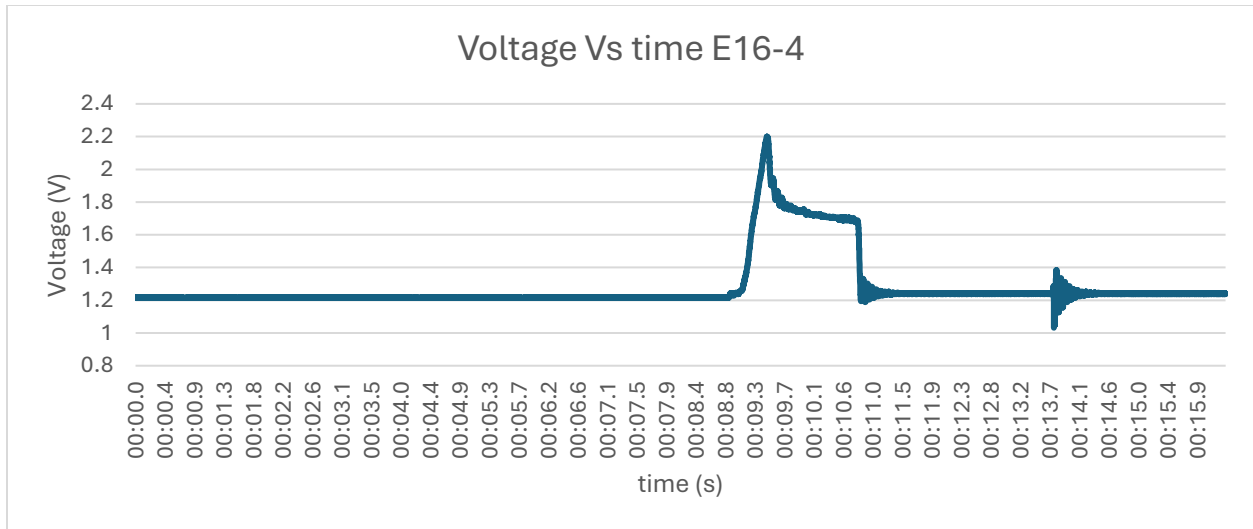


Figure 10: Voltage vs Time graph for E16-4 test

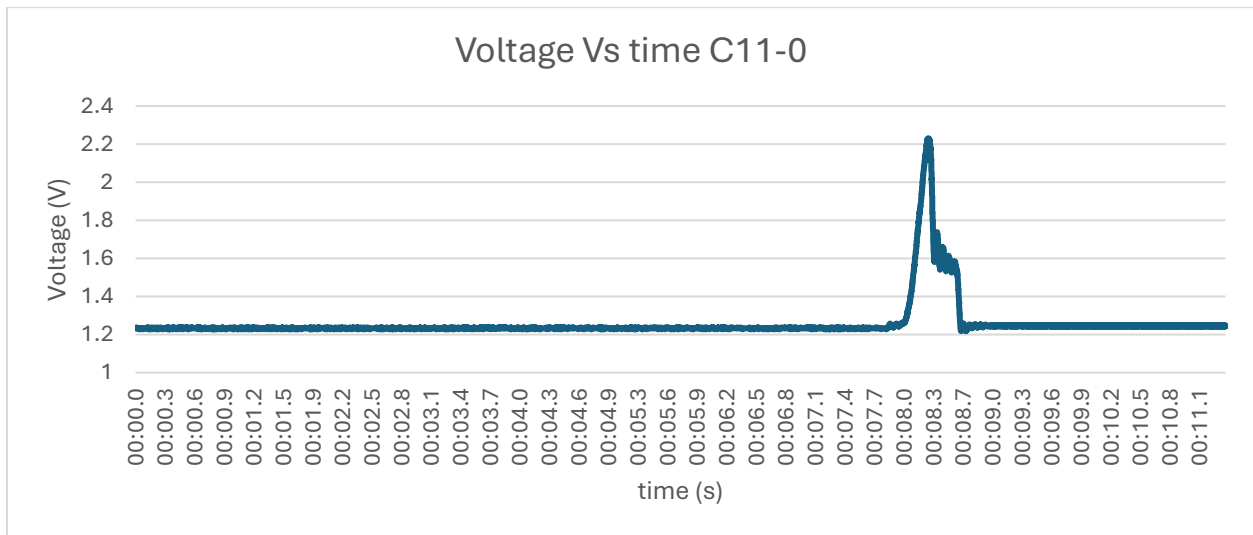


Figure 11: Voltage vs Time graph for C11-0 test

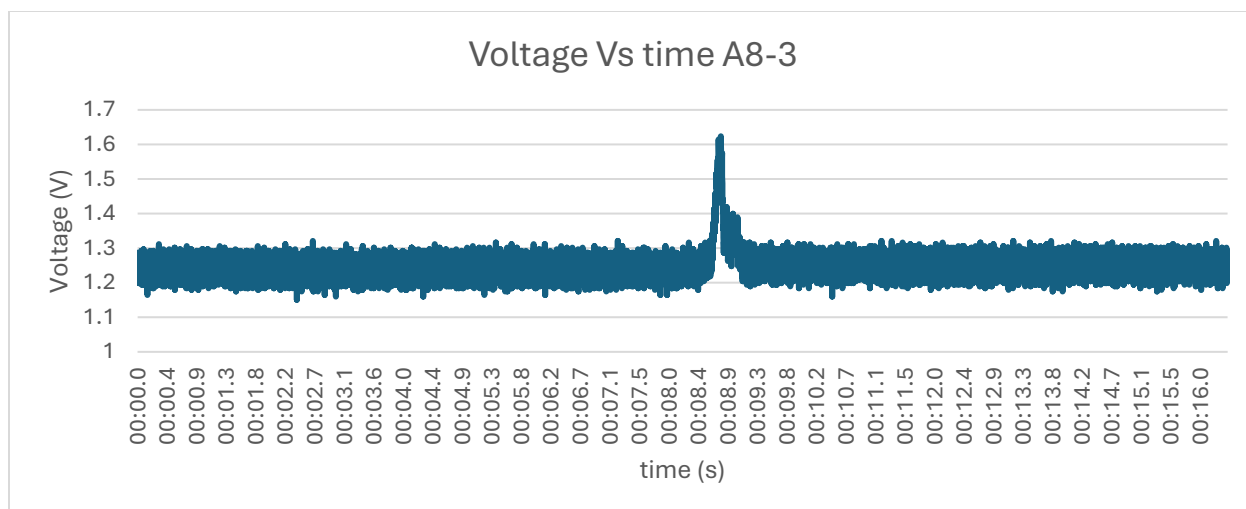


Figure 12: Voltage vs Time graph for A803 test

Analysis

To find the thrust of the rockets a calibration test needed to first be conducted. This tests values are used to find the conversion equation from voltage to force. The resulting values in the test are shown in table 1. The voltage values are an average over the time the weight was paced on the device.

Mass	Force	Voltage
0	0	1.24316
1	-9.81	0.939518
2	-19.62	0.640217
3	-29.43	0.351554

These values were then used through multiple the system of equations to find the average conversion equation found in equation 1.

$$F = 32.931 * V - 40.732 \quad (1)$$

To verify this equation the voltages in the calibration can be converted and graphed seen in figure 13.

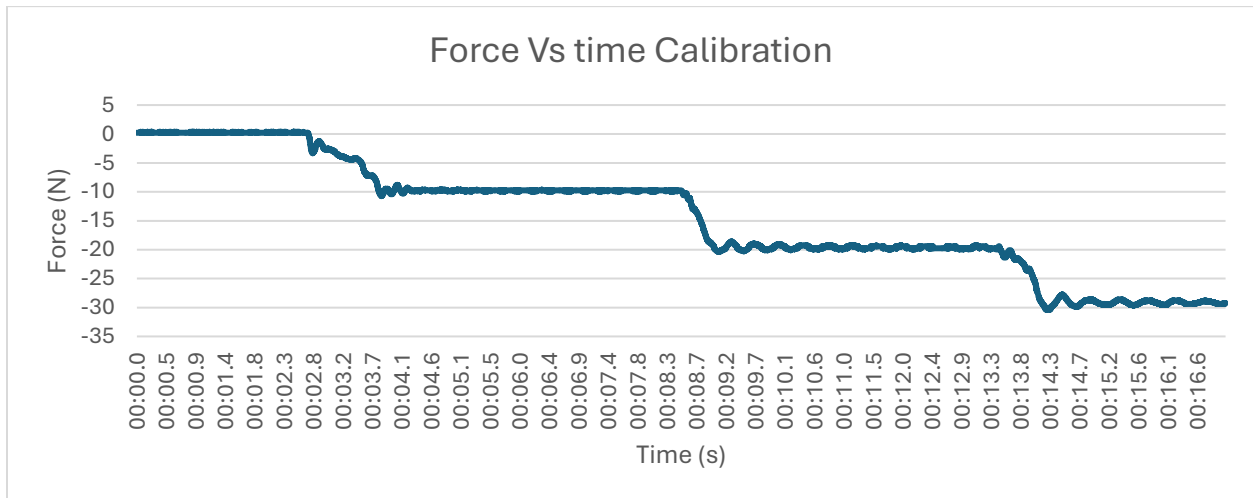


Figure 13: Force vs time graph for calibration test.

Equation 1 is then used on all tests to get figures 14 through 18.

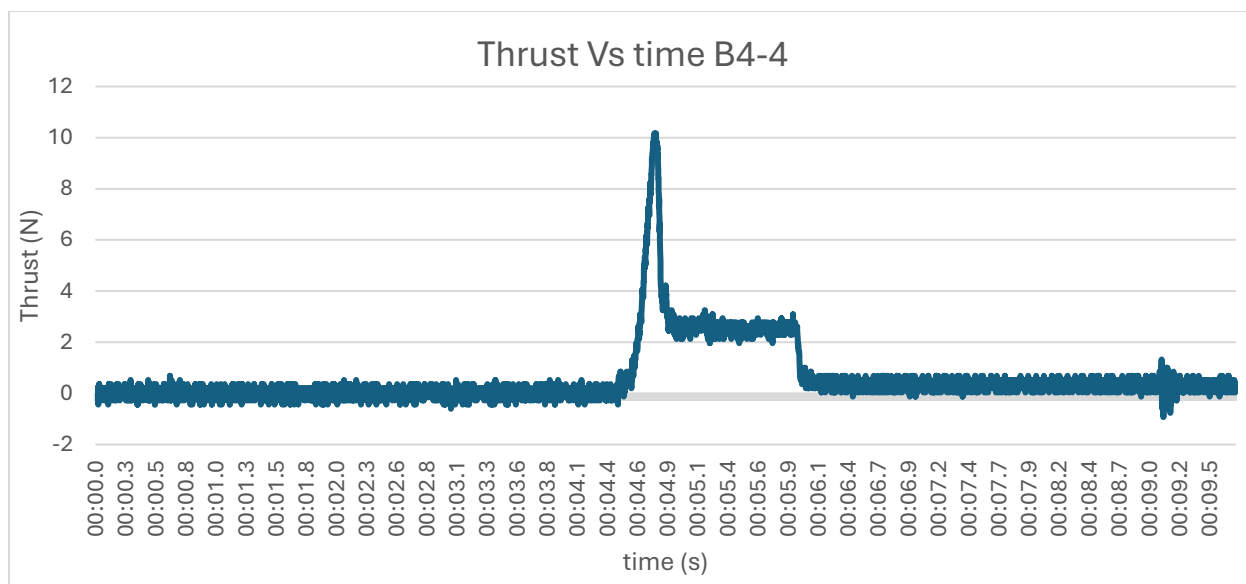


Figure 14: Thrust to time graph for B4-4 test

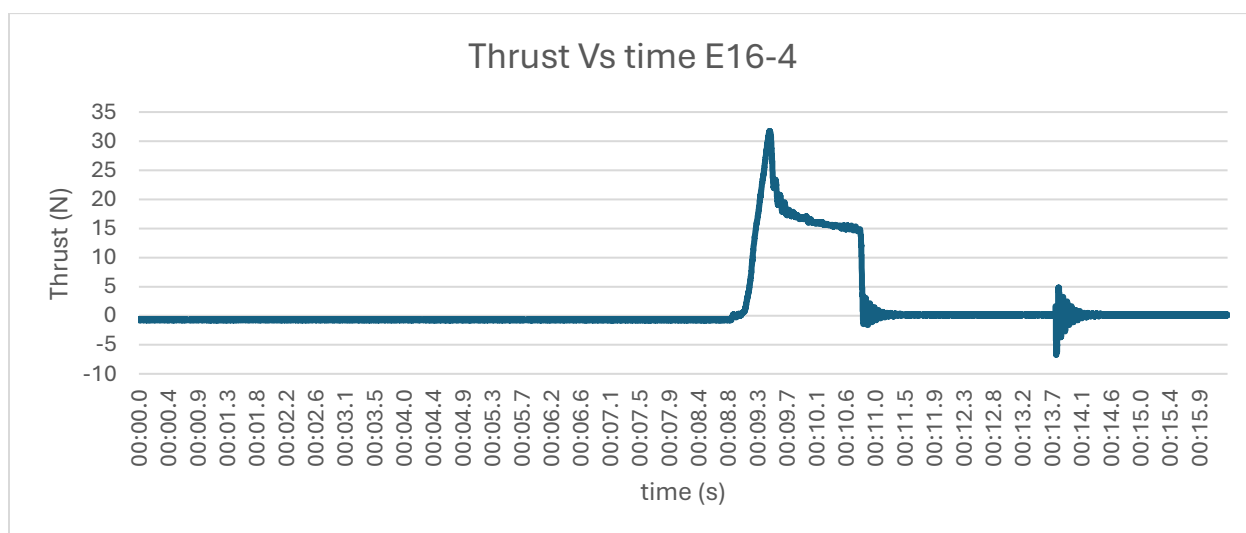


Figure 15: Thrust to time graph for E16-4 test

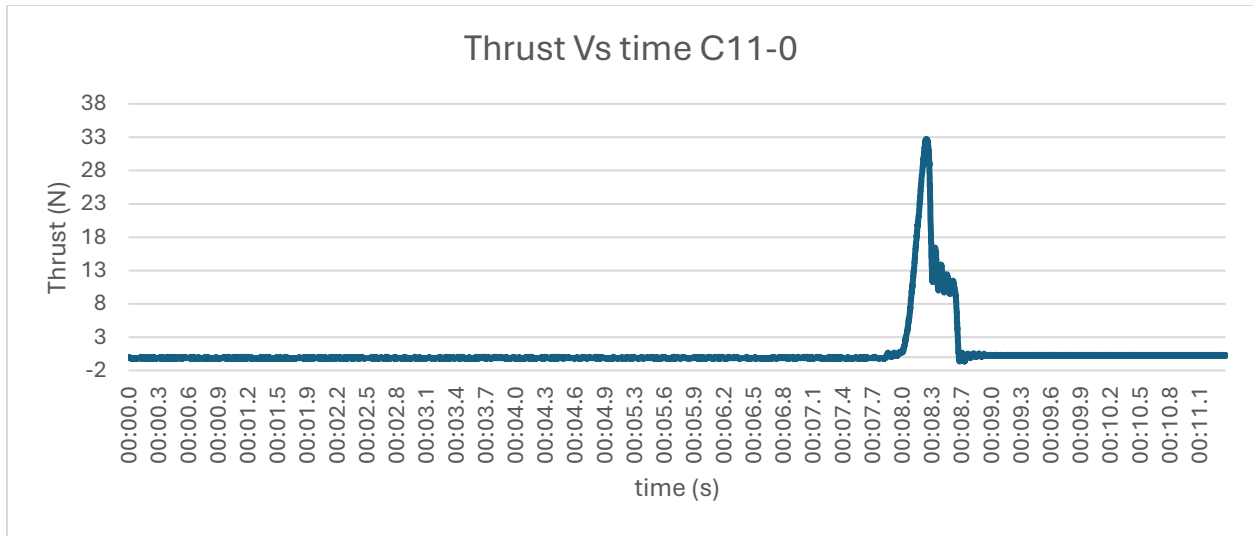


Figure 16: Thrust to time graph for C11-0 test

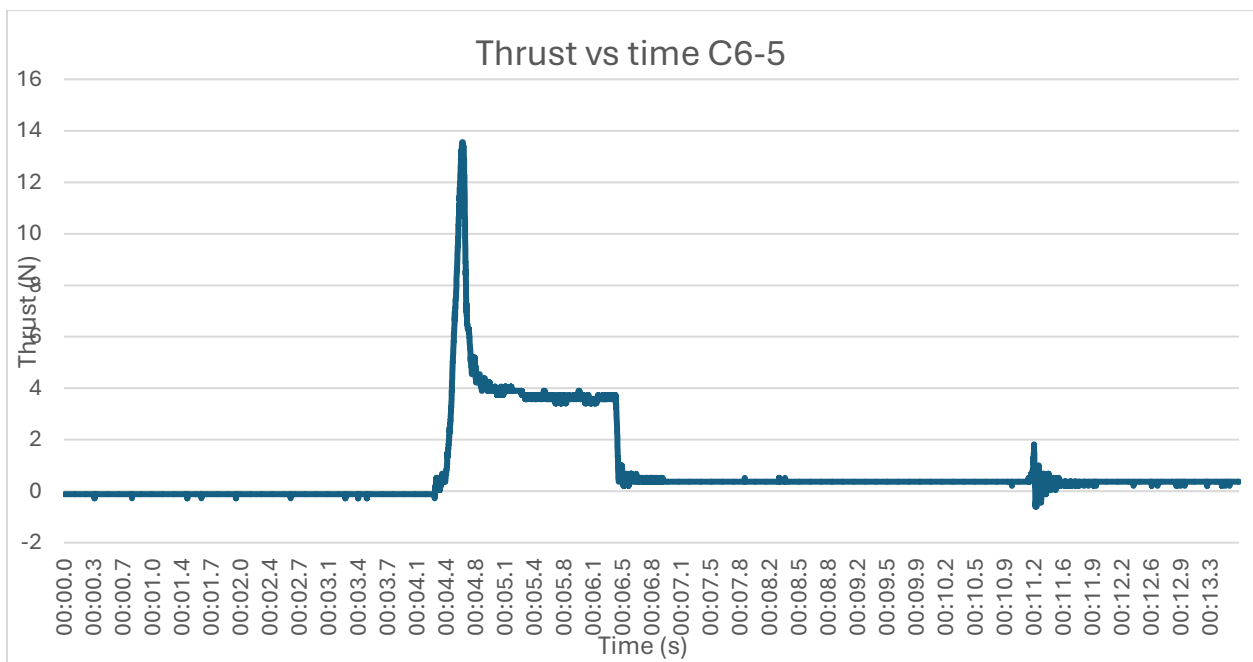


Figure 17: Thrust to time graph for C6-5 test

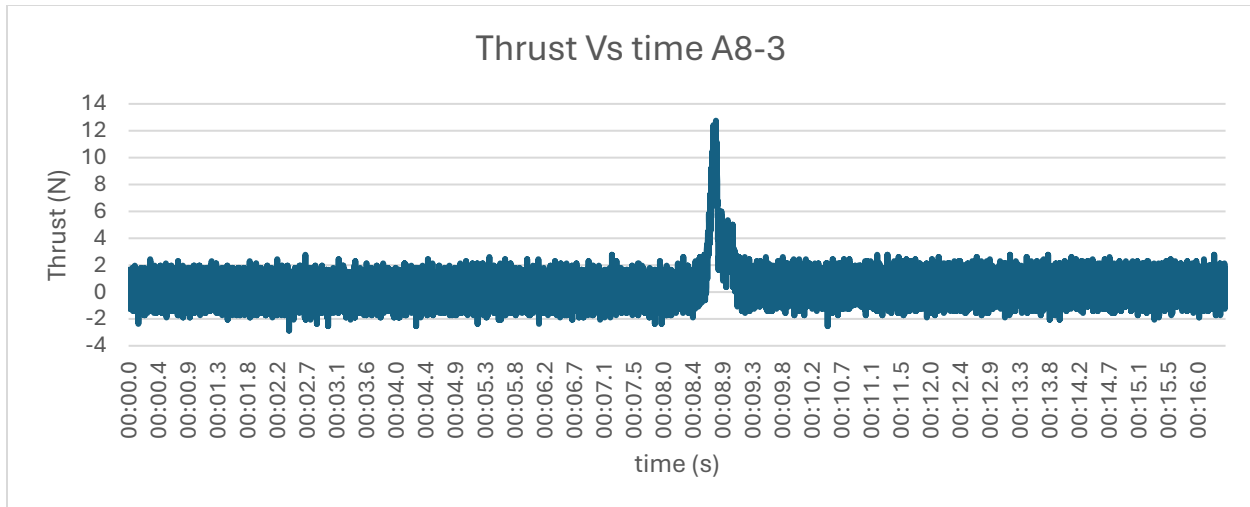


Figure 18: Thrust to time graph for A8-3 test

In these tests motor C6-5 was selected to analyze further. This resulted in the following values.

Table 2: Test C6-5 values

Max thrust (N)	13.551
Constant thrust (N)	3.731
Total impulse (N-sec)	8.526
Delay time (sec)	4.8
Burn time (sec)	2.1
Average thrust (N)	4.6

To gather some of these values equations 2 and 3 are used. While the rest were gathered through analysis of the graph.

$$I_t = \int_0^{t_b} F(t) dt \quad (2)$$

$$\bar{F} = \frac{I_t}{t_b} \quad (3)$$

Where:

I_t : total impulse

t_b : burn time

$F(t)$: thrust

$\bar{F}$: average thrust

The max thrust, total impulse, delay time and burn time can be compared to manufacture specs seen in table 3. These then produce the percent differences provided in table 4

Table 3: Estes C6-5 values

Max thrust (N)	15.3
burn time (sec)	1.6
Total impulse (N-sec)	10
delay time (sec)	5

Table 4: Percent difference between Estes and test

Max thrust (N)	12.12097811
burn time (sec)	27.03
Total impulse (N-sec)	15.91062008
delay time (sec)	4.081632653

Most of these values are in a relatively acceptable range of 15%, though the burn time was longer by 27% than the listed time. The total impulse is just outside of 15%. These altered values could result in the rocket not acting as expected, thus not reaching a desired altitude or mass requirement.

Conclusion

The analysis shows that for the Estes C6-5 Solid Rocket Motor, while relatively similar, had significant deviations in parameters when compared to those provided by the manufacturer. Looking towards the percent difference previously calculated in Table 4, the percent differences range from as low as 4% all the way to 27%. While 4% and upwards of possibly 15% can be deemed acceptable, 27% is a massive deviation that could cause issues when calculating desired outcomes. Some of the 27% difference could be attributed to possible complications with the construction of the motor itself or possible degradation over time not unlike what happens in aerospace, parts and mechanisms degrade over time, causing drops in performance or deviations from the expected output. A specific example as shown from this experiment is when comparing the maximum thrust values between the reported and the experimental value, Tables 3 and 2 going from an expected 15.3 N to 13.5 N. While not significant for smaller motors such as those used, this deviation could increase with higher output engines. A simple way to reduce or narrow down the possible causes of error is repeating the test using a fresh motor.

Acknowledgements

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A. Bharadwaj: Abstract and Method

J. Nguyen Jr.: Conclusion

R. Orr: Introduction and Analysis

K. Satvaldy: Equipment and Results

References

D. J. Syneck, *AE 416/417 Aerospace Structures and Instrumentation Laboratory Bulletins and Supplemental Information*, Embry-Riddle Aeronautical University, Daytona Beach, FL (2024).

Rabin Thapa. Evolution in Propellant of the Rocket Engine. *Journal of Materials Physics and Chemistry*. Vol. 11, No. 1, 2023, pp 8-21. <https://pubs.sciepub.com/jmpc/11/1/2>